

Rigidity of the midpoint mean value property on intervals of fixed length

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Abstract

The mean value theorem produces an interior point where the derivative equals the average rate of change, but says nothing about where that point lies. A classical converse states that if the *midpoint* works for *every* interval, the function is a polynomial of degree at most two. We ask what survives when the hypothesis is weakened to a single fixed interval length. The classification changes: those polynomials are no longer the only solutions, and an infinite family of exponentially growing oscillations appears. Rigidity is nevertheless recovered under a growth restriction, and we determine the sharp threshold. Writing $\alpha_1 = 5.5373565660 \dots$ for twice the real part of the first zero of $\sinh z - z$ in the open first quadrant, we prove: every solution of $f(a+1) - f(a) = f'(a + \frac{1}{2})$ satisfying $|f(x)| \leq Me^{cx}$ with $c < \alpha_1$ is a polynomial of degree at most two, and the constant α_1 cannot be increased. More generally we prove a complete expansion theorem, expressing every solution of at most exponential growth as a finite combination of explicit exponential and exponential-oscillatory modes, and we show that some growth hypothesis is unavoidable by constructing solutions of super-exponential growth.

Introduction

The mean value theorem is a statement about existence, not location. If f is continuous on $[a, b]$ and differentiable on (a, b) , then

$$f(b) - f(a) = (b - a)f'(\xi)$$

for some $\xi \in (a, b)$, and nothing in the theorem says where ξ is. A natural question is which functions make the answer as simple as possible, namely $\xi = \frac{a+b}{2}$.

For quadratic polynomials this always works, by a one-line computation. The converse is classical: if the midpoint is a mean value point of *every* interval, then f is a polynomial of degree at most 2. See Richmond and Richmond [8] for this and related characterizations of parabolas, and Aczél [1] and Haruki [6] for the functional-equation version. A weighted form of the location question, where the average is

taken against a weight, is treated in [11]. We reprove the classical result in Section 3; the proof is two lines, and it sets up the expectation that the rest of this paper breaks.

That classical hypothesis is a large one. It quantifies over all pairs (a, b) at once, so it is no surprise that it pins f down. This paper asks what survives when only one interval length is kept.

Question 1. *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and suppose that*

$f(a + 1) - f(a) = f'(a + \frac{1}{2})$ for every $a \in \mathbb{R}$, so that the midpoint is a mean value point of every interval of length one. Must f be a polynomial of degree at most two?

Almost everyone I have asked answers “parabolas again,” and so did I at first. The classical theorem trains that intuition, and it does not survive the weaker hypothesis. The answer is no. There is an infinite family of explicit counterexamples, the first of which is

$$u_1(x) = e^{\alpha_1 x} \cos(\beta_1 x), \alpha_1 = 5.5373565660 \dots, \beta_1 = 14.9953525556 \dots$$

a fast-growing oscillation completing one cycle about every 0.419 units. It solves (1) and it is not a parabola.

The point is not that counterexamples exist but that they are expensive: every non-polynomial solution of (1) must grow at least as fast as $e^{\alpha_1 |x|}$. That is the first main result.

Theorem 2 (Rigidity below the threshold). *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and satisfy (1). If there are constants $M > 0$ and $0 \leq c < \alpha_1 = 5.5373565660 \dots$ with $|f(x)| \leq M e^{c|x|}$ for all $x \in \mathbb{R}$, then f is a polynomial of degree at most 2. The constant α_1 is sharp: the function u_1 above satisfies (1) and*

$$|u_1(x)| \leq e^{\alpha_1 |x|}, \text{ and it is not a polynomial.}$$

The number α_1 is not arbitrary. Substituting $f(x) = e^{\lambda x}$ into (1) reduces the problem to

$$\sinh z = z, z = \lambda/2,$$

whose zero set consists of a triple zero at the origin, accounting for the classical solutions $1, x, x^2$, together with a countable family of simple complex zeros. The constant α_1 is twice the smallest positive real part occurring among those complex zeros.

This is the first level of a full classification. As the growth budget passes α_1 , then α_2 , and so on, new modes become available, and every solution within a budget is a finite combination of the modes that fit inside it.

Theorem 3 (Expansion theorem). *Let f satisfy (1) and $|f(x)| \leq Me^{c|x|}$ for some $c \geq 0$. Then there is a real polynomial p of degree at most 2 and real constants A_k, B_k, C_k, D_k such that*

$$f(x) = p(x) + \sum_{k: \alpha_k \leq c} [e^{\alpha_k x} (A_k \cos \beta_k x + B_k \sin \beta_k x) + e^{-\alpha_k x} (C_k \cos \beta_k x + D_k \sin \beta_k x)], \text{ where } \alpha_k + i\beta_k \text{ runs over}$$

the characteristic exponents described in Theorem 12. The sum is finite, and it is empty precisely when

$$c < \alpha_1.$$

The growth hypothesis cannot simply be dropped: in Section 8 we construct solutions of (1) that grow faster than any exponential, so the exponential classes do not exhaust the solution space.

Why the question is not routine

Three features distinguish Question 1 from an exercise.

First, the obvious guess is wrong, for a reason invisible from the classical proof: that proof differentiates in the interval length, which is exactly the freedom a fixed length removes.

Second, the counterexamples are explicit rather than abstract. They are closed-form oscillating exponentials, and the governing constant can be computed to any precision; we report twelve significant figures in Section 5 and check (1) for u_1 to fifty digits in Section 7.

Third, the equation is of mixed type. Written symmetrically, (1) becomes

$$f'(x) = f\left(x + \frac{1}{2}\right) - f\left(x - \frac{1}{2}\right),$$

so the derivative at a point is tied to values both ahead of it and behind it. Advanced-retarded equations admit no initial value problem in the usual sense: one cannot march forward from data on an interval, and the ordinary differential equation toolkit does not apply. See [9, 7] for the general theory. The structure has to be recovered by transform methods instead.

How the proof works

The proofs of Theorems 2 and 3 use a two-sided Laplace transform. A solution may grow on both sides, so there is no strip of common convergence and the bilateral transform converges nowhere. The way around this is to transform the two half-lines separately. The resulting one-sided transforms, defined in disjoint half-planes, turn out to be Φ and $-\Phi$ for a single meromorphic function

$$\Phi = P/\Delta,$$

where Δ is the characteristic function and P is an entire function built from the values of f on the single interval $[-\frac{1}{2}, \frac{1}{2})$ (Section 6). Inverting each transform and sliding the contour across the strip $|Re s| \leq c$

produces the finitely many modes that fit inside the growth budget, plus an error that decays exponentially at both ends. That error is itself a solution of (1), and its bilateral transform *does* converge, on a genuine strip, where it must vanish. That closes the argument.

The root analysis in Section 5 is elementary and self-contained. Separating $\sinh z = z$ into real and imaginary parts gives two real equations, which combine into a single monotone relation $y = Y(x)$ between the coordinates of a zero together with the identity $\tanh y/y = \tanh x/x$. Monotonicity of Y localizes every zero to an explicit interval, and the identity gives existence and uniqueness in each one. Together they prove, with no numerical input, that the growth rates α_k are strictly increasing, so α_1 is the binding constraint.

Setup

Definition 4. Let $h > 0$. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ has the midpoint mean value property at scale h if f is differentiable on $\mathbb{R}$ and $f(a + h) - f(a) = hf'(a + \frac{h}{2})$ for all $a \in \mathbb{R}$. We write M_h for the set of such f .

Note that (2) presupposes that f' exists at every point, since a ranges over all of $\mathbb{R}$. Substituting $a = x - \frac{h}{2}$ puts (2) in the symmetric form

$$f'(x) = \frac{1}{h} [f(x + \frac{h}{2}) - f(x - \frac{h}{2})], x \in \mathbb{R},$$

which is the form we shall use throughout. For $h = 1$ we abbreviate $M := M_1$ and (3) reads

$$f'(x) = f(x + \frac{1}{2}) - f(x - \frac{1}{2}).$$

Theorem 5 (Scaling). For $h > 0$ let $(T_h f)(x) := f(hx)$. Then $f \in M_h$ if and only if $T_h f \in M_1$. Moreover $|f(x)| \leq Me^{c|x|}$ for all x if and only if $|(T_h f)(x)| \leq Me^{ch|x|}$ for all x .

Proof. Write $g = T_h f$, so $g'(x) = hf'(hx)$. Then $g(x + \frac{1}{2}) - g(x - \frac{1}{2}) = f(hx + \frac{h}{2}) - f(hx - \frac{h}{2})$,

which by (3) equals $hf'(hx) = g'(x)$ exactly when (3) holds at the point hx . Since $x \mapsto hx$ is a bijection of $\mathbb{R}$, the two conditions are equivalent. The growth statement is immediate. $\square$

So nothing is lost by fixing $h = 1$: every statement below transfers to scale h by replacing each growth constant c with c/h . The threshold at scale h is α_1/h , as expected, since shorter intervals impose the condition more often and should be more rigid.

Theorem 6 (Automatic smoothness). Every $f \in M$ belongs to $C^\infty(\cdot)$.

Proof. A differentiable function is continuous, so the right-hand side of (4) is continuous, hence f' is continuous and $f \in C^1$. If $f \in C^n$ then the right-hand side of (4) is C^n , so $f' \in C^n$ and $f \in C^{n+1}$. Induction completes the proof. $\square$

It is convenient to name the operator on the right of (4). For $u: \mathbb{R} \rightarrow \mathbb{R}$ put

$$(Su)(x) := u\left(x + \frac{1}{2}\right) - u\left(x - \frac{1}{2}\right),$$

so that (4) reads $f' = Sf$. Since S commutes with differentiation, every $f \in M$ satisfies $f^{(n)} = S^n f$, and expanding the binomial gives

$$f^{(n)}(x) = \sum_{j=0}^n (-1)^j \binom{n}{j} f\left(x + \frac{n-2j}{2}\right).$$

Theorem 7 (Solutions of exponential growth are entire). *Let $f \in M$ satisfy $|f(x)| \leq Me^{c|x|}$ on $\mathbb{R}$ for some $M > 0$, $c \geq 0$. Then f extends to an entire function of exponential type at most $A := 2e^{c/2}$, and*

$$|f(x + iy)| \leq Me^{c|x|} e^{A|y|} \quad (x, y \in \mathbb{R}).$$

Proof. Every point occurring on the right of (6) lies within distance $n/2$ of x , so

$$|f^{(n)}(x)| \leq 2^n \sup_{|t-x| \leq n/2} |f(t)| \leq 2^n Me^{c(|x|+n/2)} = Me^{c|x|} A^n.$$

In particular $|f^{(n)}(0)| \leq MA^n$, so the power series $F(w) := \sum_{n \geq 0} f^{(n)}(0)w^n/n!$ has infinite radius of convergence and defines an entire function. On the real axis the Lagrange form of the remainder gives, for real t ,

$$|f(t) - \sum_{n < N} \frac{f^{(n)}(0)}{n!} t^n| \leq \frac{|t|^N}{N!} \sup_{|\xi| \leq |t|} |f^{(N)}(\xi)| \leq Me^{c|t|} \frac{(A|t|)^N}{N!} \rightarrow 0,$$

so $F = f$ on $\mathbb{R}$, and F is an entire extension of f . Being entire and equal to f on $\mathbb{R}$, the function F has Taylor coefficients $f^{(n)}(x_0)/n!$ at every real x_0 , whence for all $w \in \mathbb{C}$

$$|F(x_0 + w)| \leq \sum_{n \geq 0} \frac{|f^{(n)}(x_0)|}{n!} |w|^n \leq Me^{c|x_0|} e^{A|w|}.$$

Taking $w = iy$ gives the stated bound. $\square$

Functions of this kind, bounded by $Me^{c|x|}$ on the real axis and of exponential type in the imaginary direction, form a classical class; see Boas [3, Ch. 6]. Theorem 7 is not needed later, but it explains why the exponential ansatz of Section 4 is not a guess: among exponentially bounded solutions there are no non-analytic objects to look for.

The classical case

We record the theorem that trains the wrong intuition.

Theorem 8 (Classical). *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and suppose $f \in M_h$ for every $h > 0$. Then f is a polynomial of degree at most 2.*

Proof. By hypothesis $f \in M_1$, so $f \in C^\infty$ by Theorem 6. Rewriting (2) in the symmetric form (3) and multiplying by h , we have

$$f\left(t + \frac{h}{2}\right) - f\left(t - \frac{h}{2}\right) = hf'(t) \text{ for all } t \in \mathbb{R}, h > 0,$$

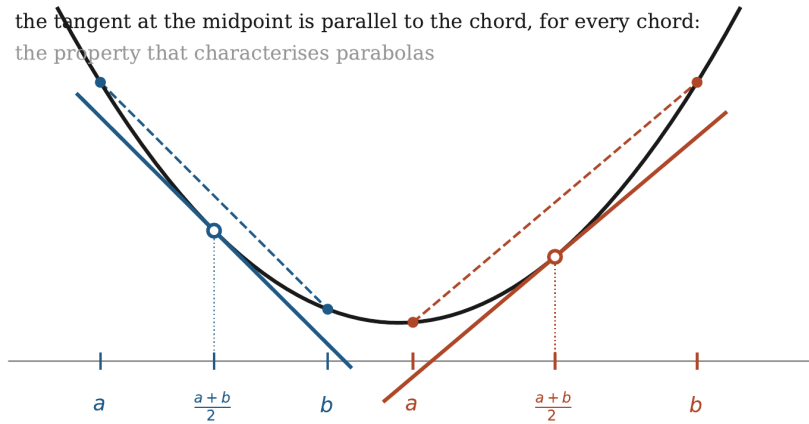
and by antisymmetry the identity holds for all $h \in \mathbb{R}$. Both sides are smooth in h . Differentiating twice with respect to h gives

$$\frac{1}{4} [f''\left(t + \frac{h}{2}\right) - f''\left(t - \frac{h}{2}\right)] = 0 \text{ for all } t, h,$$

so f'' is constant and f is a polynomial of degree at most 2. The converse is the computation

$$\left[\left(t + \frac{h}{2}\right)^2 - \left(t - \frac{h}{2}\right)^2\right]' = 2th = h \cdot (t^2)'. \quad \square$$

The proof differentiates in h , so fixing h destroys exactly this step and Theorem 8 says nothing about M_1 on its own. Figure 1 shows the classical situation.



The classical picture. For a parabola, the chord over $[a, b]$ is parallel to the tangent at the midpoint, for every choice of $a < b$. Theorem 8 says that no other differentiable function does this. The present paper asks what happens when only one value of $b - a$ is used.

The characteristic equation

Substituting $f(x) = e^{\lambda x}$, $\lambda \in \mathbb{C}$, into (4) and dividing by $e^{\lambda x}$ gives

$$\lambda = e^{\lambda/2} - e^{-\lambda/2} = 2\sinh\left(\frac{\lambda}{2}\right).$$

Accordingly we define the *characteristic function*

$$\Delta(\lambda) := \lambda - 2\sinh\left(\frac{\lambda}{2}\right), \Lambda := \{\lambda \in \mathbb{C} : \Delta(\lambda) = 0\},$$

so that $e^{\lambda x} \in M \otimes \mathbb{C}$ if and only if $\lambda \in \Lambda$. Substituting $z = \lambda/2$ turns $\Delta(\lambda) = 0$ into the clean form

$$\sinh z = z$$

Functions such as Δ , polynomials in λ and $e^{\mu\lambda}$, are the *exponential polynomials* that serve as characteristic functions of all linear autonomous difference-differential equations; Bellman and Cooke [2, Ch. 12] is the standard reference for the distribution of their zeros. We write

$$E(z) := \sinh z - z = \frac{z^3}{6} + \frac{z^5}{120} + \frac{z^7}{5040} + \dots,$$

so that $\Delta(\lambda) = -2E(\lambda/2)$ and $\Lambda = 2E^{-1}(0)$.

Two consequences are immediate from (9). First, $z = 0$ is a zero of E of multiplicity exactly three, since the coefficient of z^3 is $1/6 \neq 0$. Second, the corresponding solutions of (4) are 1 , x and x^2 , obtained in the usual way by differentiating $e^{\lambda x}$ with respect to λ at $\lambda = 0$. So the parabolas are still there. They are simply no longer alone: the following section shows that E has infinitely many further zeros.

We also record the expansion of Δ at the origin, which will be used to identify the residue there:

$$\Delta(\lambda) = -\frac{\lambda^3}{24} - \frac{\lambda^5}{1920} - \dots.$$

The zeros

Everything in Theorem 12 is proved without numerical input; the values in Table 1 only evaluate quantities the theorem has already localized.

Throughout, $z = x + iy$ with $x, y \in \mathbb{R}$. Separating $\sinh z = z$ into real and imaginary parts using $\sinh(x + iy) = \sinh x \cosh y + i \cosh x \sinh y$ gives the pair

$$\sinh x \cosh y = x, \cosh x \sinh y = y.$$

Theorem 9. *E is odd with real Taylor coefficients, so its zero set is invariant under $z \mapsto -z$ and $z \mapsto z'$. Moreover E has no nonzero real zero and no nonzero purely imaginary zero, and every nonzero zero of E is simple.*

Proof. The symmetries are clear from (9). On the real axis $\sinh x > x$ for $x > 0$, and on the imaginary axis $E(iy) = i(\sin y - y)$, which vanishes only at $y = 0$. For simplicity, note $E'(z) = \cosh z - 1$, which vanishes exactly at $z = 2\pi i n$, $n \in \mathbb{Z}$; and $E(2\pi i n) = -2\pi i n \neq 0$ for $n \neq 0$. Hence no zero of E other than $z = 0$ is a zero of E' . $\square$

By Theorem 9 it suffices to locate the zeros in the open first quadrant $x > 0$, $y > 0$.

Theorem 10 (Band localization). *Let $z = x + iy$ be a zero of E with $x > 0$ and $y > 0$. Then $\cos y > 0$, $\sin y > 0$, $\frac{\tan y}{y} = \frac{\tanh x}{x}$, and consequently $y \in J_k := \left(2\pi k, 2\pi k + \frac{\pi}{2}\right)$ for some integer $k \geq 1$.*

Proof. Since $\sinh x > x > 0$, the first equation of (11) gives $\cos y = x/\sinh x \in (0, 1)$; the second gives $\sin y = y/\cosh x > 0$. So y lies in the open first quadrant modulo 2π , that is, $y \in J_k$ for some integer $k \geq 0$. Dividing the second equation of (11) by the first,

$$\coth x \tan y = \frac{y}{x} \Leftrightarrow \frac{\tan y}{y} = \frac{\tanh x}{x}.$$

Now $\tanh x/x < 1$ for every $x > 0$, while $\tan y/y > 1$ for $y \in (0, \pi/2)$. Hence $k = 0$ is impossible. $\square$

The next result is the engine of the section.

Theorem 11 (The coordinate relation). *Define $Y(x) := \coth x \sqrt{\sinh^2 x - x^2} = \cosh x \sqrt{1 - \frac{x^2}{\sinh^2 x}}$, $x > 0$.*

Then Y is a strictly increasing bijection from $(0, \infty)$ onto $(0, \infty)$, and every zero $z = x + iy$ of E with $x, y > 0$ satisfies $y = Y(x)$.

Proof. That Y is real and positive follows from $\sinh x > x > 0$. Squaring and adding the two equations (11) gives $x^2/\sinh^2 x + y^2/\cosh^2 x = 1$, whence $y = Y(x)$ for $y > 0$; this is the last assertion.

For monotonicity, differentiate $Y^2 = \cosh^2 x - x^2 \coth^2 x$:

$$(Y^2)'(x) = 2\sinh x \cosh x - \frac{2x \cosh^2 x}{\sinh^2 x} + \frac{2x^2 \cosh x}{\sinh^3 x} = \frac{2 \cosh x}{\sinh^3 x} Q(x),$$

where

$$Q(x) := \sinh^4 x - x \sinh x \cosh x + x^2.$$

We claim $Q > 0$ on $(0, \infty)$, which gives $(Y^2)' > 0$ and hence the strict monotonicity of Y .

Regard Q as a quadratic in x with leading coefficient 1. Its discriminant is

$$D(x) = \sinh^2 x \cosh^2 x - 4 \sinh^4 x = \sinh^2 x (\cosh^2 x - 4 \sinh^2 x),$$

which is negative as soon as $\tanh x > \frac{1}{2}$, that is for $x > \operatorname{artanh} \frac{1}{2} = \frac{1}{2} \log 3$. For such x we get $Q(x) > 0$ immediately.

For $0 < x \leq \frac{1}{2} \log 3$ we argue directly. Since $\sinh x \geq x$ we have $\sinh^4 x \geq x^4$, and $x \sinh x \cosh x - x^2 = \frac{x}{2} (\sinh 2x - 2x)$, so with $u := 2x \in (0, \log 3]$,

$$Q(x) \geq x^4 - \frac{x}{2} (\sinh u - u) = \frac{x}{2} [2x^3 - (\sinh u - u)] = \frac{x}{2} \left[\frac{u^3}{12} - \sum_{j \geq 2} \frac{u^{2j+1}}{(2j+1)!} \right],$$

using $2x^3 = u^3/4$ and $\sinh u - u = u^3/6 + \sum_{j \geq 2} u^{2j+1}/(2j+1)!$. For $0 < u \leq \log 3 < 1.1$,

$$\sum_{j \geq 2} \frac{u^{2j+1}}{(2j+1)!} \leq \frac{u^5}{120} \sum_{i \geq 0} \left(\frac{u^2}{42} \right)^i \leq \frac{u^5}{120} \cdot \frac{1}{1 - \frac{1.21}{42}} < \frac{u^5}{116},$$

and $u^3/12 > u^5/116$ holds whenever $u^2 < 116/12$, in particular for $u \leq 1.1$. Hence $Q > 0$ on $(0, \frac{1}{2} \log 3]$ as well.

Finally $Y(0^+) = 0$, since $\sinh^2 x - x^2 = \frac{1}{3}x^4 + O(x^6)$ gives $Y(x) \sim x/\sqrt{3}$, and $Y(x) \rightarrow \infty$ as $x \rightarrow \infty$

because $Y(x) \geq \sqrt{\sinh^2 x - x^2} \rightarrow \infty$. Strict monotonicity and continuity then make Y a bijection onto $(0, \infty)$. $\square$

Theorem 12 (Structure of the zero set). *Let $E(z) = \sinh z - z$ and $\Delta(\lambda) = \lambda - 2 \sinh(\lambda/2)$.*

1. *$z = 0$ is a zero of E of multiplicity exactly 3; correspondingly $\lambda = 0$ is a zero of Δ of multiplicity exactly 3, and the associated solutions of (4) are $1, x, x^2$.*
2. *For each integer $k \geq 1$ there is exactly one zero $z_k = x_k + iy_k$ of E with $x_k > 0$ and $y_k \in J_k = \left(2\pi k, 2\pi k + \frac{\pi}{2}\right)$, and these, together with 0 and the images of the z_k under $z \mapsto -z$ and $z \mapsto z'$, exhaust the zero set of E . All nonzero zeros are simple.*
3. *$x_k \in I_k := \left(Y^{-1}(2\pi k), Y^{-1}\left(2\pi k + \frac{\pi}{2}\right)\right)$, the intervals I_k are pairwise disjoint and increasing, and consequently $x_1 < x_2 < x_3 < \dots$. In particular $x_1 > Y^{-1}(2\pi) = 2.60660 \dots$ and $x_k \rightarrow \infty$.*
4. *$x_k = \log(4\pi k) + O(1/k)$ and $y_k = 2\pi k + \frac{\pi}{2} + O\left(\frac{\log k}{k}\right)$ as $k \rightarrow \infty$.*

Writing $\lambda_k := 2z_k = \alpha_k + i\beta_k$, the zero set of Δ is $\Lambda = \{0\} \cup \{\pm \lambda_k, \pm \overline{\lambda_k} : k \geq 1\}$, the sequence $\alpha_k = 2x_k$ is strictly increasing with $\alpha_k \rightarrow \infty$, and $\alpha_k = 2\log(4\pi k) + O(1/k)$.

Proof. (i) is (9) and (10), together with the standard fact that a zero of multiplicity m of the characteristic function contributes the solutions $e^{\lambda x}, xe^{\lambda x}, \dots, x^{m-1}e^{\lambda x}$; for $\lambda = 0$, $m = 3$, these are $1, x, x^2$, and one checks directly that each satisfies (4).

For (ii), fix $k \geq 1$ and parametrize the band by $y \in J_k$, setting $x := Y^{-1}(y) > 0$, which is legitimate by Theorem 11. We claim that $z = x + iy$ is a zero of E if and only if

$$A(y) := \frac{\tanh y}{y} = \frac{\tanh(Y^{-1}(y))}{Y^{-1}(y)} =: B(y).$$

Indeed, set $p = \cos y$, $q = \sin y$, $P = x/\sinh x$, $Q = y/\cosh x$. The relation $y = Y(x)$ says exactly $P^2 + Q^2 = 1$, and of course $p^2 + q^2 = 1$; the relation (12) says exactly $qP = pQ$. Two unit vectors with vanishing cross product are equal up to sign, so $(p, q) = \pm (P, Q)$; and since $y \in J_k$ gives $p = \cos y > 0$ while $x > 0$ gives $P > 0$, the sign is $+$. Thus $(p, q) = (P, Q)$, which is precisely the pair (11). The converse implication is Theorems 10 and 11.

Now compare A and B on J_k . The function A is strictly increasing on J_k , from $A(2\pi k^+) = 0$ to

$$A\left(\left(2\pi k + \frac{\pi}{2}\right)^-\right) = +\infty: \text{ indeed}$$

$$A'(y) = \frac{y \sec^2 y - \tanh y}{y^2} = \frac{\sec^2 y}{y^2} \left(y - \frac{1}{2} \sin 2y \right) > 0 \text{ on } J_k,$$

since $y > 2\pi > \frac{1}{2} \geq \frac{1}{2} \sin 2y$ there. The function $t \mapsto \tanh t/t$ is strictly decreasing and positive on $(0, \infty)$, and Y^{-1} is strictly increasing, so B is strictly decreasing and positive on J_k , with a finite positive value at each endpoint. Hence $A - B$ is strictly increasing on J_k , negative near the left endpoint and $+\infty$ at the right endpoint, so it has exactly one zero there. This gives exactly one zero z_k of E per band. Theorems 9 and 10 guarantee that these, their conjugates and negatives, and 0, are all of the zeros. Simplicity is Theorem 9.

(iii) Since $y_k = Y(x_k) \in J_k$ and Y is an increasing bijection, $x_k \in I_k$. As $2\pi(k+1) > 2\pi k + \frac{\pi}{2}$, the intervals I_k are disjoint and increasing, so x_k is strictly increasing. Since $y_k > 2\pi k \rightarrow \infty$ and $y_k \leq \cosh x_k$ by (11), $x_k \rightarrow \infty$.

(iv) Rewrite $\sinh z = z$ as $e^z = 2z + e^{-z}$. At $z = z_k$ we have $x_k \geq x_1 > 2.6$, hence $|e^{-z_k}| < e^{-2.6} < 0.08$, while $|2z_k| \geq 2y_k > 4\pi k$. Therefore

$$e^{z_k} = 2z_k(1 + O(k^{-1})),$$

and taking real parts of the logarithm, $x_k = \log|2z_k| + O(k^{-1})$, while taking imaginary parts gives $y_k = \arg(2z_k) + 2\pi k$ for the appropriate branch. Since $\arg(2z_k) = \frac{\pi}{2} - \arctan(x_k/y_k) = \frac{\pi}{2} + O(x_k/k)$ and $x_k = O(\log k)$ by the previous display, we obtain $y_k = 2\pi k + \frac{\pi}{2} + O(\log k/k)$. Then $|2z_k| = 2y_k(1 + O(x_k^2/y_k^2)) = 4\pi k(1 + O(1/k))$, so $x_k = \log(4\pi k) + O(1/k)$. $\square$

Remark 13. Numerically the next term in (iv) is visible: $k(x_k - \log(4\pi k))$ equals 0.2377 at $k = 1$, 0.2631 at $k = 10$, 0.2542 at $k = 100$ and 0.2509 at $k = 1000$, consistent with $x_k = \log(4\pi k) + \frac{1}{4k} + o(1/k)$. We do not need this refinement.

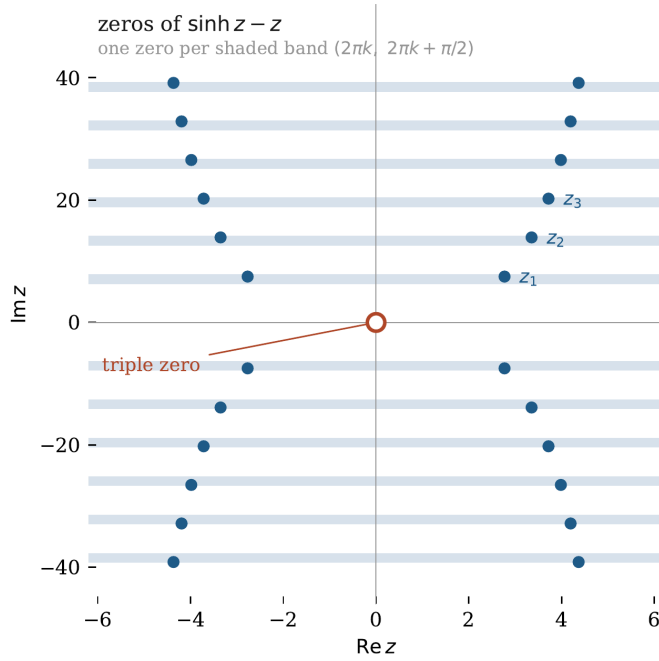
Remark 14 (Relation to the Lambert W function). The identity $e^z = 2z + e^{-z}$ is a perturbation of $e^z = 2z$, whose solutions are $z = -W_k(-\frac{1}{2})$ for the branches W_k of the Lambert W function [4]. This is the source of the asymptotics in Theorem 12(iv) and it also supplies the excellent starting values used by the root finder in Section 7.

Table 1 lists the first several zeros. Figure 2 shows their location, with the bands J_k of Theorem 10 shaded; the picture makes visible both the triple zero at the origin and the one-zero-per-band structure proved in Theorem 12(ii).

The first eight zeros of $\sinh z - z$ in the open first quadrant and the corresponding characteristic exponents λ_k of (4). All digits shown are correct; the computation is described in Section 7.

k	z_k	$\lambda_k = 2z_k$	growth rate $\alpha_k = \operatorname{Re} \lambda_k$
1	2.7686782830 + 5.5373565660 i	5.5373565660 + 11.0747131320 i	5.5373565660
2	3.3522098849 + 6.7044197697 i	6.7044197697 + 13.4088395394 i	6.7044197697
3	3.7167676798 + 7.4335353595 i	7.4335353595 + 14.8670707190 i	7.4335353595
4	3.9831416403 + 7.9662832807 i	7.9662832807 + 15.9325665614 i	7.9662832807
5	4.1932514704 + 8.3865029409 i	8.3865029409 + 16.7730058818 i	8.3865029409

k	z_k	$\lambda_k = 2z_k$	growth rate $\alpha_k = \operatorname{Re} \lambda_k$
6	4.3667951177 + 8.7335902353 i	8.7335902353 + 17.4671804706 i	8.7335902353
7	4.5146404495 + 9.0292808990 i	9.0292808990 + 18.0585617980 i	9.0292808990
8	4.6434279571 + 9.2868559141 i	9.2868559141 + 18.5737118282 i	9.2868559141



Zeros of $\sinh z - z$. The origin carries a zero of multiplicity three, corresponding to the classical solutions $1, x, x^2$. Every other zero is simple and lies in one of the shaded bands $\operatorname{Im} z \in \left(2\pi k, 2\pi k + \frac{\pi}{2}\right)$, exactly one per band (Theorem 12).

Method and main results

Throughout this section $f \in M$ satisfies

$$|f(x)| \leq M e^{c|x|} \quad (x \in \mathbb{R})$$

for some $M > 0$ and $c \geq 0$. We define

$$m := \#\{k \geq 1 : \alpha_k \leq c\},$$

which is finite by Theorem 12, and we fix once and for all

$$\sigma \in (\max(c, 0), \alpha_{m+1}), \sigma' \in (-\alpha_{m+1}, \min(-c, 0)),$$

both nonempty since $\alpha_{m+1} > c \geq 0$.

The two one-sided transforms

For $\text{Res} > c$ set $F_+(s) := \int_0^\infty f(x)e^{-sx}dx$, and for $\text{Res} < -c$ set $F_-(s) := \int_{-\infty}^0 f(x)e^{-sx}dx$; both converge absolutely by (13). Define the entire function

$$P(s) := f(0) - \int_0^{1/2} f\left(\frac{1}{2} - v\right)e^{sv}dv - \int_0^{1/2} f\left(w - \frac{1}{2}\right)e^{-sw}dw.$$

Note that P depends only on the restriction of f to the single interval $[-\frac{1}{2}, \frac{1}{2})$.

Theorem 15. *With Δ as in (7) and P as in (15),*

$\Delta(s)F_+(s) = P(s)$ ($\text{Res} > c$), $\Delta(s)F_-(s) = -P(s)$ ($\text{Res} < -c$). Consequently, with $\Phi := P/\Delta$, meromorphic on $\mathbb{C}$ with poles contained in Λ , we have $F_+ = \Phi$ on $\{\text{Res} > c\}$ and $F_- = -\Phi$ on $\{\text{Res} < -c\}$.

Proof. Let $\text{Res} > c$. Integration by parts gives $\int_0^\infty f'e^{-sx}dx = -f(0) + sF_+(s)$, the boundary term at infinity vanishing by (13). Substituting $u = x \pm \frac{1}{2}$,

$$\int_0^\infty f\left(x + \frac{1}{2}\right)e^{-sx}dx = e^{s/2}[F_+(s) - \int_0^{1/2} f(u)e^{-su}du], \quad \int_0^\infty f\left(x - \frac{1}{2}\right)e^{-sx}dx = e^{-s/2}[F_+(s) + \int_{-1/2}^0 f(u)e^{-su}du].$$

Inserting these into (4) and using $e^{s/2} - e^{-s/2} = 2\sinh(s/2)$,

$$-f(0) + sF_+(s) = 2\sinh\left(\frac{s}{2}\right)F_+(s) - e^{s/2} \int_0^{1/2} f(u)e^{-su}du - e^{-s/2} \int_{-1/2}^0 f(u)e^{-su}du.$$

Rearranging and substituting $v = \frac{1}{2} - u$ in the first integral and $w = u + \frac{1}{2}$ in the second gives

$\Delta(s)F_+(s) = P(s)$ with P as in (15). The computation for F_- is identical except that the boundary term appears with the opposite sign and the truncations are complementary; it yields $\Delta(s)F_-(s) = -P(s)$. $\square$

That a single meromorphic Φ governs both half-lines is what makes the argument work: the two half-line transforms live in disjoint half-planes, with no common strip on which a bilateral transform of f exists, yet each is the analytic continuation of the other up to sign.

Theorem 16 (Estimates on vertical lines). *Fix a compact interval $[\sigma', \sigma) \subset \mathbb{R}$. There is a constant C and a threshold $T_0 \geq 1$, depending only on the strip, such that for $s = u + it$ with $u \in [\sigma', \sigma)$ and $|t| \geq T_0$,*

$|P(s) - f(0)| \leq \frac{C}{|t|}, |\Delta(s) - s| \leq C, |\Phi(s) - \frac{f(0)}{s}| \leq \frac{C}{t^2}, |\Phi(s)| \leq \frac{C}{|t|}$, The threshold T_0 may be taken larger than the height of every pole of Φ in the strip, of which there are finitely many.

Proof. Since $|2\sinh(s/2)| \leq 2\cosh(u/2)$ is bounded on the strip, $|\Delta(s) - s| \leq C$; in particular $|\Delta(s)| \geq |t| - |u| - C \geq |t|/2$ for $|t|$ large. For P , integrate by parts once in each integral of (15): for instance

$$\int_0^{1/2} f\left(\frac{1}{2} - v\right) e^{sv} dv = \frac{1}{s} [f(0) e^{s/2} - f\left(\frac{1}{2}\right)] + \frac{1}{s} \int_0^{1/2} f'\left(\frac{1}{2} - v\right) e^{sv} dv,$$

which is $O(1/|t|)$ uniformly on the strip since $|e^{sv}| \leq e^{|u|/2}$ there; the second integral in (15) is handled identically. Hence $P(s) = f(0) + O(1/|t|)$. Finally $\Delta(s)^{-1} = s^{-1}(1 - (s - \Delta(s))/s)^{-1} = s^{-1} + O(t^{-2})$, and multiplying the two expansions gives the statements about Φ . $\square$

Inversion and contour shifting

Theorem 17. For σ as in (14), $\frac{1}{2\pi i} \int_{\sigma-iT}^{\sigma+iT} \Phi(s) e^{sx} ds = \{f(x), x > 0, 0, x < 0, \}$ and for σ' as in (14),

$$\frac{1}{2\pi i} \int_{\sigma'-iT}^{\sigma'+iT} (-\Phi(s)) e^{sx} ds = \{f(x), x < 0, 0, x > 0, \}$$

Proof. f is smooth by Theorem 6, hence locally of bounded variation, and F_+ converges absolutely for $\text{Re } s > c$. The complex inversion formula for the Laplace transform (see Widder [12, Ch. II]) applies to the function $f1_{(0,\infty)}$ and gives the first display, using $F_+ = \Phi$ on the line $\text{Re } s = \sigma > c$ from Theorem 15. The second display is the same statement applied to $f1_{(-\infty,0)}$, whose Laplace transform is $F_- = -\Phi$ on $\text{Re } s = \sigma' < -c$. $\square$

Theorem 18 (Shifting the contour). Let $q(x) := \sum_{\lambda} \text{Res}_{s=\lambda} [\Phi(s) e^{sx}]$, the sum being over the finitely many

$\lambda \in \Lambda$ with $\sigma' < \text{Re } \lambda < \sigma$. Then there is $C > 0$ with

$$|f(x) - q(x)| \leq C e^{\sigma' x} (x \geq 0), |f(x) - q(x)| \leq C e^{\sigma x} (x \leq 0).$$

Proof. By (14) and Theorem 12, the elements of Λ in the closed strip $\sigma' \leq \text{Re } s \leq \sigma$ are exactly 0 and $\pm \lambda_k, \pm \overline{\lambda_k}$ for $k \leq m$, a finite set, none of them on the boundary lines. Apply the residue theorem on the rectangle with vertical sides $\text{Re } s = \sigma', \text{Re } s = \sigma$ and horizontal sides $\text{Im } s = \pm T$. By Theorem 16 the integrand $\Phi(s) e^{sx}$ is bounded by $(C/T) \max(e^{\sigma x}, e^{\sigma' x})$ on the horizontal sides, whose length is fixed, so those contributions tend to 0 as $T \rightarrow \infty$. Hence

$$\frac{1}{2\pi i} \int_{(\sigma)} \Phi(s) e^{sx} ds - \frac{1}{2\pi i} \int_{(\sigma')} \Phi(s) e^{sx} ds = q(x),$$

where $\int_{(\cdot)}$ denotes the upward principal-value integral along the indicated vertical line.

Let $x > 0$. By Theorem 17 the first term equals $f(x)$, so $f(x) = q(x) + \frac{1}{2\pi i} \int_{(\sigma')} \Phi(s) e^{sx} ds$. Split

$\Phi(s) = f(0)/s + R(s)$. Since $\sigma' < 0$ and $x > 0$, the classical formula $\frac{1}{2\pi i} \int_{(\sigma')} \frac{e^{sx}}{s} ds = 0$ applies. By

Theorem 16, R is absolutely integrable on the line $\text{Re } s = \sigma'$, so

$$\left| \frac{1}{2\pi i} \int_{(\sigma')} \Phi(s) e^{sx} ds \right| = \left| \frac{1}{2\pi i} \int_{(\sigma')} R(s) e^{sx} ds \right| \leq \frac{e^{\sigma'x}}{2\pi} \int_{\mathbb{R}} |R(\sigma' + it)| dt = C e^{\sigma'x}.$$

For $x < 0$ the second display of Theorem 17 gives $f(x) = -\frac{1}{2\pi i} \int_{(\sigma')} \Phi(s) e^{sx} ds = q(x) - \frac{1}{2\pi i} \int_{(\sigma)} \Phi(s) e^{sx} ds$,

and now $\sigma > 0$, $x < 0$ give $\frac{1}{2\pi i} \int_{(\sigma)} \frac{e^{sx}}{s} ds = 0$ and the same bound with σ in place of σ' . The estimates

extend to $x = 0$ by continuity after enlarging C . $\square$

Theorem 19. *The function q of Theorem 18 has the form*

$q(x) = p(x) + \sum_{k=1}^m [e^{\alpha_k x} (A_k \cos \beta_k x + B_k \sin \beta_k x) + e^{-\alpha_k x} (C_k \cos \beta_k x + D_k \sin \beta_k x)]$ with p a real polynomial of degree at most 2 and $A_k, B_k, C_k, D_k \in \mathbb{R}$. In particular $q \in M$, and q is a polynomial of degree at most 2 when $m = 0$.

Proof. By (10), Δ has a zero of order 3 at the origin, so $\Phi(s) e^{sx}$ has a pole of order at most 3 there,

and its residue is $\frac{1}{2} \frac{d^2}{ds^2} [s^3 \Phi(s) e^{sx}]|_{s=0}$, a polynomial in x of degree at most 2. At each nonzero $\lambda \in \Lambda$

the pole is simple by Theorem 12, with residue $\frac{P(\lambda)}{\Delta'(\lambda)} e^{\lambda x}$. Both P and Δ are real on the real axis, hence

$\overline{\Phi(s')} = \Phi(s)$, so the residues at λ and λ' are complex conjugates; grouping λ_k with $\overline{\lambda_k}$, and $-\lambda_k$ with

$-\overline{\lambda_k}$, produces real coefficients in the stated form. Each summand is a real solution of (4), being a real

linear combination of $e^{\lambda x}$ with $\Delta(\lambda) = 0$, and $1, x, x^2 \in M$, so $q \in M$. $\square$

Theorem 20 (Expansion theorem). *Let $f \in M$ satisfy $|f(x)| \leq M e^{c|x|}$ on $\mathbb{R}$ for some $M > 0$, $c \geq 0$. Then $f = q$, where q is the function of Theorem 19. Explicitly, there are a real polynomial p of degree at*

most 2 and real constants A_k, B_k, C_k, D_k such that

$$f(x) = p(x) + \sum_{k: \alpha_k \leq c} [e^{\alpha_k x} (A_k \cos \beta_k x + B_k \sin \beta_k x) + e^{-\alpha_k x} (C_k \cos \beta_k x + D_k \sin \beta_k x)].$$

Proof. Set $u := f - q$. By Theorem 19, $u \in M$, and by Theorem 18,

$$|u(x)| \leq C e^{-\delta|x|} \text{ with } \delta := \min(\sigma, -\sigma') > 0.$$

In particular $u \in L^1(\mathbb{R})$ and the bilateral transform $U(s) := \int_{\mathbb{R}} u(x) e^{-sx} dx$ converges absolutely and is

analytic on the strip $|\operatorname{Re} s| < \delta$. Because u and u' decay exponentially, integration by parts has no boundary terms and the shift substitutions produce no truncation terms:

$$\int_{\mathbb{R}} u'(x) e^{-sx} dx = sU(s), \int_{\mathbb{R}} u\left(x \pm \frac{1}{2}\right) e^{-sx} dx = e^{\pm s/2} U(s).$$

Applying these to $u' = Su$ gives $sU(s) = 2 \sinh(s/2) U(s)$, that is $\Delta(s)U(s) = 0$ on the strip. Since $\Delta \not\equiv 0$, its zeros are isolated, so U vanishes on the strip minus a discrete set and hence, by continuity, identically. Taking $s = it$ shows that the Fourier transform of the integrable function u vanishes identically, so $u = 0$ almost everywhere, and $u = 0$ everywhere by continuity. Therefore $f = q$, which is the asserted formula by Theorem 19, since by (14) the indices k appearing there are exactly those with $\alpha_k \leq c$. $\square$

The rigidity theorem of the introduction now follows, with its sharpness.

Theorem 21 (Rigidity and sharpness). *Let $c_* := \alpha_1 = 5.5373565660 \dots$*

1. *If $f \in M$ and $|f(x)| \leq M e^{c|x|}$ on $\mathbb{R}$ for some $M > 0$ and $0 \leq c < c_*$, then f is a polynomial of degree at most 2.*
2. *The function $u_1(x) = e^{\alpha_1 x} \cos(\beta_1 x)$ belongs to M , satisfies $|u_1(x)| \leq e^{c_*|x|}$ for all x , and is not a polynomial. Hence the constant c_* in (i) cannot be replaced by any larger one.*

Proof. (i) With $c < \alpha_1$ we have $m = 0$, so the sum in Theorem 20 is empty and $f = p$.

(ii) $\lambda_1 = \alpha_1 + i\beta_1 \in \Lambda$ by Theorem 12, so $e^{\lambda_1 x}$ solves the complexification of (4); taking real parts,

$u_1 \in M$. For $x \geq 0$, $|u_1(x)| \leq e^{\alpha_1 x} = e^{c_*|x|}$, and for $x \leq 0$, $|u_1(x)| \leq e^{\alpha_1 x} \leq 1 \leq e^{c_*|x|}$. Finally u_1 is unbounded and oscillates infinitely often, so it is not a polynomial. If (i) held with some $c' > c_*$ in place

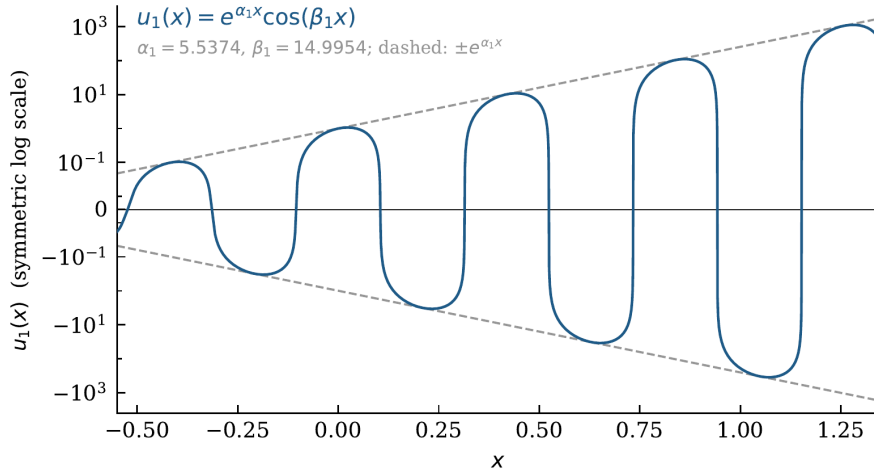
of c_* , then u_1 , which satisfies the growth bound with $c = c_* < c'$, would have to be a polynomial, a contradiction. $\square$

Remark 22 (The bound must be two-sided). *The hypothesis in Theorem 21 is $|f(x)| \leq Me^{c|x|}$ on all of $\mathbb{R}$, and the absolute value in the exponent is essential. The one-sided condition $f(x) = O(e^{cx})$ as $x \rightarrow +\infty$ does not suffice for any c , however small. Indeed Λ is symmetric under $\lambda \mapsto -\lambda$, so $v_1(x) := e^{-\alpha_1 x} \cos(\beta_1 x) \in M$ as well, and v_1 is bounded on $[0, \infty)$, hence $O(e^{cx})$ as $x \rightarrow +\infty$ for every $c \geq 0$, while being neither a polynomial nor small as $x \rightarrow -\infty$. This symmetry is forced: the map $f \mapsto f(-\cdot)$ sends M to itself, as one checks directly from (4).*

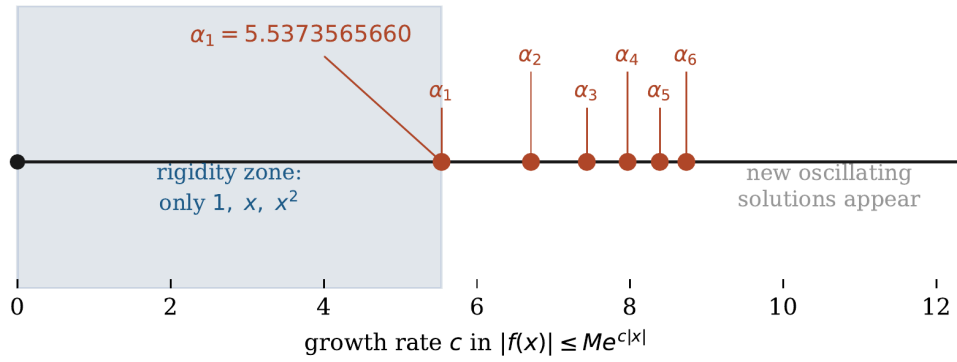
Theorem 23 (General scale). *Fix $h > 0$. If $f \in M_h$ and $|f(x)| \leq Me^{c|x|}$ on $\mathbb{R}$ with $0 \leq c < c_*/h$, then f is a polynomial of degree at most 2, and the constant c_*/h is sharp. More generally, the expansion of Theorem 20 holds at scale h with α_k, β_k replaced by $\alpha_k/h, \beta_k/h$.*

Proof. Immediate from Theorem 5, Theorem 20 and Theorem 21. $\square$

Figure 3 plots the first oscillating solution against its envelope, and Figure 4 summarizes the whole classification on a single growth-rate axis.



The first oscillating solution $u_1(x) = e^{\alpha_1 x} \cos(\beta_1 x)$, shown on a symmetric logarithmic scale so that the oscillation and the exponential growth are visible at once. The dashed lines are the envelope $\pm e^{\alpha_1 x}$. The period is $2\pi/\beta_1 = 0.4190088418 \dots$



The classification of Theorem 20 on one axis. Below the threshold $\alpha_1 = 5.5373565660$ the only solutions of (1) are $1, x$ and x^2 . Each subsequent characteristic exponent α_k admits four further real solutions, $e^{\pm \alpha_k x} \cos \beta_k x$ and $e^{\pm \alpha_k x} \sin \beta_k x$.

Numerical verification

All computations were carried out in mpmath at 50 decimal digits of working precision. The accompanying supplement contains a script that regenerates every number in this paper.

Root finding

Starting values were generated from the fixed-point iteration $z \mapsto \log(2z) + 2\pi i k$ suggested by the form $e^z = 2z + e^{-z}$ (equivalently, from the Lambert W branches, cf. [4]); these were refined by Newton's method for $E(z) = \sinh z - z$, whose derivative $\cosh z - 1$ is bounded away from zero at every nonzero root by Theorem 9. The residuals satisfy $|E(z_k)| < 10^{-48}$ for all $k \leq 40$, and the values obtained at 50 and at 80 digits of working precision agree to more than 50 digits. Independently, a numerical evaluation of the argument principle

$$\frac{1}{2\pi i} \oint_{\partial R} \frac{E'(z)}{E(z)} dz$$

returns 3 for the rectangle $R = [-0.5, 2.6) \times [-9, 9)$, confirming that the only zero with $0 < \operatorname{Re} z \leq 2.6$ would have to lie outside that box, which by the a priori bound $|\operatorname{Im} z| \leq \cosh(\operatorname{Re} z) \leq \cosh(2.6) < 9$ of (11) is impossible; and it returns 1 for each of the boxes $[2.6, 2.82) \times [6.2, 7.9)$ and $[3.2, 3.5) \times [12.5, 14.2)$, in agreement with Theorem 12(ii) and with $I_1 \subset (2.6066, 2.8117)$.

The identity, checked directly

Let $u_1(x) = e^{\alpha_1 x} \cos(\beta_1 x)$ and define the normalized residual

$$\rho(x) := e^{-\alpha_1 x} |u_1(x+1) - u_1(x) - u_1'(x + \frac{1}{2})|.$$

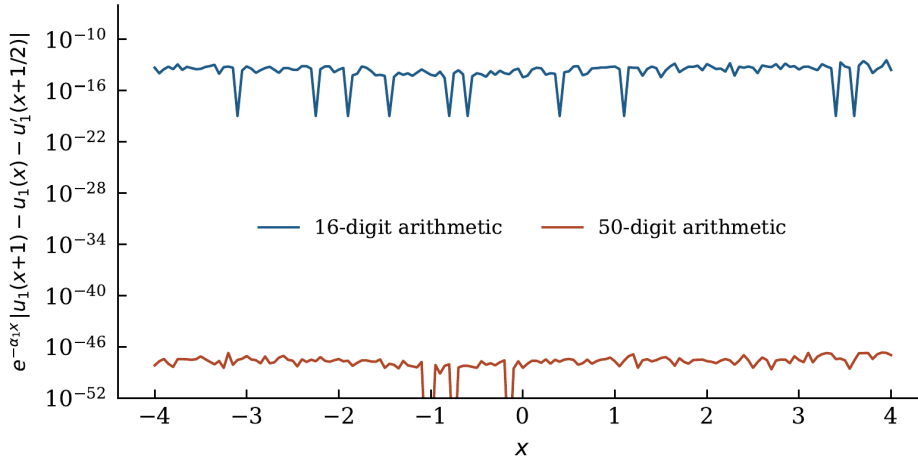
The normalization removes the exponential factor common to all three terms, so ρ measures the failure of (1) relative to the size of the solution. Sampling x at spacing 0.05 on $[-4, 4]$ gives

$$\max \rho \approx 3.8 \times 10^{-13} (16 - \text{digit arithmetic}), \max \rho \approx 3.8 \times 10^{-47} (50 - \text{digit arithmetic}).$$

The residual tracks the working precision rather than saturating, which is the signature of an exact identity evaluated in floating point rather than an approximate relation. The loss of three to four digits relative to full precision is accounted for by the size of the individual terms relative to the normalization:

$e^{-\alpha_1 x} u_1(x+1)$ and $e^{-\alpha_1 x} u_1'(x + \frac{1}{2})$ have magnitudes up to $e^{\alpha_1} \approx 254$ and $e^{\alpha_1/2} |\lambda_1| \approx 255$, and the phase $\beta_1 x$ reaches about 60 over the sampling range, so rounding is amplified by a factor of order

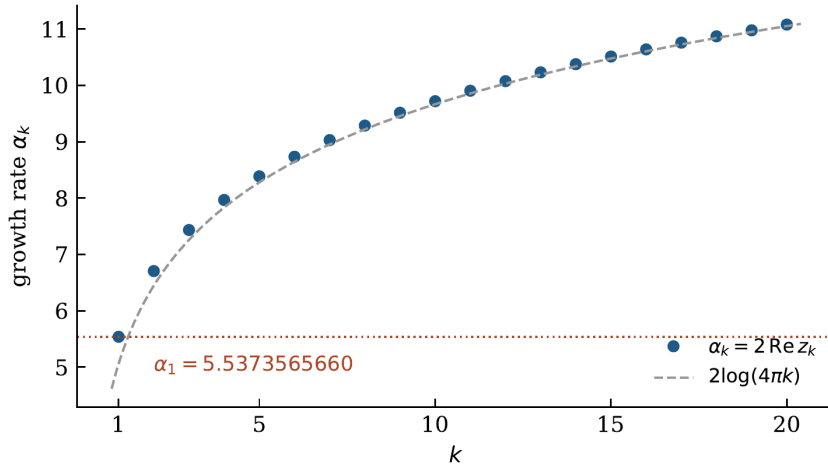
$e^{\alpha_1} \beta_1 \approx 3.8 \times 10^3$. Measuring that amplification directly, $\max \rho$ divided by 10^{-d} at d digits of working precision equals 3.8×10^3 , 2.3×10^3 , 1.5×10^3 and 3.8×10^3 for $d = 16, 25, 35, 50$: bounded and independent of d , as it must be for an exact identity. Figure 5 shows ρ at both precisions.



The normalized residual ρ of the midpoint identity for u_1 , evaluated at two working precisions. The level of each curve is set by the arithmetic, not by the function: the identity is exact.

Monotonicity of the growth rates

Theorem 12(iii) proves α_k strictly increasing, so no numerical check is required, but the computed values agree. Figure 6 shows them against the asymptotic $2\log(4\pi k)$ of Theorem 12(iv); the strict increase was verified for $k \leq 2000$.



The growth rates $\alpha_k = 2\operatorname{Re} z_k$ against k , with the asymptotic $2\log(4\pi k)$ of Theorem 12(iv). The sequence is strictly increasing, so α_1 is the binding constraint in Theorem 21.

Discussion

Why there is no initial value problem

Equation (4) ties $f'(x)$ to values of f both ahead of x and behind it. It is therefore an autonomous linear functional differential equation of *mixed type*, also called advanced-retarded. For a retarded equation such as $f'(x) = f(x - 1)$ one prescribes f on an interval of length 1 and integrates forward, and the solution space is parametrized by that data. For a mixed equation neither direction is available, and the difference is not cosmetic. The general theory, developed by Rustichini [9, 10] and Mallet-Paret [7], and standing in deliberate contrast to the retarded theory of Hale and Verduyn Lunel [5], replaces the initial value problem with a Fredholm theory on the whole line and an exponential dichotomy governed by the location of the characteristic roots; our Theorem 20 is an explicit, elementary instance of that picture for one particular equation.

Two contrasts make the point.

First, for a linear ordinary differential equation of order n the solution space has dimension n , and the solution is determined by n numbers at a point. Here the solution space is infinite dimensional, containing

the independent modes of Theorem 12 and, by Theorem 24, a great deal more; yet no finite set of numbers at a point determines a solution, and no interval of data can be prescribed freely: by Theorem 7, a solution of exponential growth is entire, so its restriction to any interval, however short, determines it completely. The data one may prescribe is thus over-determined in one sense and under-determined in the other.

Second, the parameter P in Theorem 15 depends only on $f|_{[-1/2, 1/2]}$, and it is this single entire function that carries all the information about f . But P is not free: for $\Phi = P/\Delta$ to be the transform of a function of exponential type c , the numerator P must vanish at every $\lambda \in \Lambda$ with $|\operatorname{Re} \lambda| > c$, which is an infinite family of linear constraints on $f|_{[-1/2, 1/2]}$. Rigidity is exactly the statement that when $c < \alpha_1$ these constraints leave only a three-dimensional space.

The growth hypothesis is not removable

Theorem 20 classifies solutions of at most exponential growth. That restriction is not an artifact of the method: solutions of arbitrarily fast growth exist.

Theorem 24. *There exists $f \in M$ such that for every $c > 0$, $\limsup_{x \rightarrow +\infty} |f(x)|e^{-cx} = +\infty$. In particular f*

satisfies no bound of the form (13) and M is not exhausted by the functions listed in Theorem 20.

Proof. Since $\alpha_k \rightarrow \infty$, choose indices $k_1 < k_2 < \dots$ with $a_m := \alpha_{k_m} \geq m + 1$, and write $b_m := \beta_{k_m}$ and

$u_m(x) := e^{a_m x} \cos(b_m x)$, so that $u_m \in M$ and $|u_m'(x)| \leq (a_m + b_m)e^{a_m x}$. Put $\kappa_m := 1 + a_m + b_m$.

We choose $c_m > 0$ and $R_m > 0$ recursively. Set $c_1 := 1$ and $R_0 := 0$. Suppose $c_1, \dots, c_m$ and $R_1, \dots, R_{m-1}$ are chosen. Since $a_j < a_m$ for $j < m$, the quantity

$$\left(\sum_{j < m} c_j \kappa_j e^{a_j(R+1)} \right) / \left(c_m e^{a_m R} \right)$$

tends to 0 as $R \rightarrow \infty$; choose $R_m > R_{m-1} + m$ so large that

$$\sum_{j < m} c_j \kappa_j e^{a_j(R_m+1)} \leq \frac{1}{4} c_m e^{a_m R_m} \text{ and } c_m e^{a_m R_m} \geq 4.$$

Then put $\varepsilon_m := \frac{1}{4} c_m e^{a_m R_m}$ and choose $c_{m+1} > 0$ so small that

$$c_{m+1} \kappa_{m+1} e^{a_{m+1}(R_j+1)} \leq 2^{-(m+1)} \varepsilon_j \quad (1 \leq j \leq m), \quad c_{m+1} \kappa_{m+1} e^{a_{m+1}(m+1)} \leq 2^{-(m+1)}.$$

The second condition in (17) makes $\sum_m c_m \kappa_m e^{a_m^N}$ finite for every N , so the series

$$f(x) := \sum_{m \geq 1} c_m u_m(x)$$

and the series of first derivatives converge uniformly on every compact subset of $\mathbb{R}$. Hence f is differentiable with $f' = \sum c_m u_m'$, and since each u_m satisfies (4) and the shifted series converge uniformly on compacta as well, $f \in M$.

Fix m and choose $x_m \in [R_m, R_m + 1)$ with $\cos(b_m x_m) = 1$, which is possible because $2\pi/b_m \leq 2\pi/\beta_1 < 1$. Using the first part of (16) for the terms $j < m$ and the first part of (17) for the terms $j > m$,

$$|f(x_m)| \geq c_m e^{a_m x_m} - \sum_{j < m} c_j \kappa_j e^{a_j(R_m+1)} - \sum_{j > m} c_j \kappa_j e^{a_j(R_m+1)}$$

Since $a_m \geq m + 1$ and $c_m e^{R_m} \geq 4$ by (16),

$$|f(x_m)| \geq \frac{1}{2} c_m e^{(m+1)R_m} = \frac{1}{2} \left(c_m e^{R_m} \right) e^{mR_m} \geq 2e^{mR_m} \geq 2e^{m(x_m-1)}.$$

Given $c > 0$, take any $m > c + 1$; then $|f(x_m)| e^{-cx_m} \geq 2e^{(m-c)x_m - m} \rightarrow \infty$ along the subsequence, since $x_m \geq R_m \geq m$. This proves the claim. $\square$

Remark 25. *The construction is flexible, and variants of it produce faster growth; we do not pursue this. What matters here is only that the exponential classes of Theorem 20 form a strictly increasing hierarchy that does not exhaust M , so that a growth hypothesis of some kind is genuinely needed and is not an artifact of the transform method.*

Conclusion and open questions

Non-midpoint interior points

Fix $\theta \in (0, 1)$ and ask for the differentiable f with $f(a + 1) - f(a) = f'(a + \theta)$ for all a . The exponential ansatz now produces the characteristic function

$$G_\theta(\lambda) := e^\lambda - 1 - \lambda e^{\theta\lambda} = \sum_{n \geq 2} \left(1 - n\theta^{n-1}\right) \frac{\lambda^n}{n!},$$

and $G_{1/2}(\lambda) = -e^{\lambda/2}\Delta(\lambda)$, so $\theta = \frac{1}{2}$ recovers the equation studied here. Write $c_*(\theta)$ for the corresponding threshold, that is, the least value of $|\operatorname{Re}\lambda|$ over the nonzero zeros of G_θ ; Theorem 12 says $c_*(\frac{1}{2}) = \alpha_1$.

It is tempting to expect c_* to vary smoothly, and to guess that $\theta = \frac{1}{2}$, being the most symmetric choice, is where it is smallest. Both guesses are wrong.

Theorem 26.

1. $G_{1-\theta}(\lambda) = -e^\lambda G_\theta(-\lambda)$; consequently $c_*(\theta) = c_*(1 - \theta)$.
2. $\lambda = 0$ is a zero of G_θ of multiplicity 3 if $\theta = \frac{1}{2}$, and of multiplicity exactly 2 otherwise; in the latter case the polynomial solutions are only 1 and x .
3. There are $\eta > 0$ and a real-analytic odd function $\varepsilon \mapsto \lambda(\varepsilon)$ on $(-\eta, \eta)$ such that, writing $\varepsilon = \theta - \frac{1}{2}$, the number $\lambda_\theta := \lambda(\varepsilon)$ is for every $\varepsilon \neq 0$ a simple nonzero real zero of G_θ , and

$$\lambda_\theta = 24\left(\theta - \frac{1}{2}\right) + O\left(\left(\theta - \frac{1}{2}\right)^3\right) \text{ as } \theta \rightarrow \frac{1}{2}.$$

In particular $c_*(\theta) \leq |\lambda_\theta| \rightarrow 0$ as $\theta \rightarrow \frac{1}{2}$, while $c_*(\frac{1}{2}) = 5.5373565660$. So c_* has a jump discontinuity at $\theta = \frac{1}{2}$, and $\theta = \frac{1}{2}$ is a strict local maximum of c_* , not a minimum.

Proof. (i) Expanding, $-e^\lambda G_\theta(-\lambda) = -e^\lambda(e^{-\lambda} - 1 + \lambda e^{-\theta\lambda}) = e^\lambda - 1 - \lambda e^{(1-\theta)\lambda} = G_{1-\theta}(\lambda)$, and $\lambda \mapsto -\lambda$ preserves $|\operatorname{Re}\lambda|$.

(ii) In (18) the coefficient of λ vanishes identically, that of λ^2 is $\frac{1}{2} - \theta$, and that of λ^3 is $\frac{1}{6} - \frac{\theta^2}{2}$, which equals $\frac{1}{24} \neq 0$ at $\theta = \frac{1}{2}$.

(iii) Set $H(\lambda, \varepsilon) = G_{1/2+\varepsilon}(\lambda)/\lambda^2$, which by (18) is an entire function of two variables with real Taylor coefficients. From (ii), $H(0, 0) = 0$, $\partial_\lambda H(0, 0) = \frac{1}{24}$ and $\partial_\varepsilon H(0, 0) = -1$. The analytic implicit function theorem supplies $\eta > 0$ and a unique analytic $\lambda(\varepsilon)$ on $(-\eta, \eta)$ with $H(\lambda(\varepsilon), \varepsilon) = 0$ and $\lambda(0) = 0$, satisfying $\lambda'(0) = -\partial_\varepsilon H(0, 0)/\partial_\lambda H(0, 0) = 24$. It is real valued for real ε : H has real coefficients, so $\overline{\lambda(\varepsilon)}$ is also a zero, and uniqueness in a small disc forces $\overline{\lambda(\varepsilon)} = \lambda(\varepsilon)$. The zero is simple since $\partial_\lambda H \neq 0$ near $(0, 0)$, and $\lambda(\varepsilon) \neq 0$ for $0 < |\varepsilon| < \eta$ after shrinking η , since $\lambda(\varepsilon) = 24\varepsilon + O(\varepsilon^2)$. Finally λ is odd: by (i) the zero sets of $G_{1/2+\varepsilon}$ and $G_{1/2-\varepsilon}$ are negatives of one another, so $-\lambda(-\varepsilon)$ is a zero of $G_{1/2+\varepsilon}$ near 0, and uniqueness gives $\lambda(\varepsilon) = -\lambda(-\varepsilon)$. Hence the Taylor expansion of λ contains only odd powers and the

second-order term is absent. Since λ_θ is a nonzero zero of G_θ , the function $e^{\lambda_\theta x}$ is a non-polynomial solution obeying $|e^{\lambda_\theta x}| \leq e^{|\lambda_\theta| |x|}$, so $c_*(\theta) \leq |\lambda_\theta|$. $\square$

Numerically the next coefficient is $\lambda_\theta = 24\varepsilon + \frac{576}{5}\varepsilon^3 + O(\varepsilon^5)$; for example $\lambda_{0.45} = -1.21459145$ against $24\varepsilon = -1.2$.

The mechanism is the loss of a symmetry. At $\theta = \frac{1}{2}$ the characteristic function is odd, $\Delta(-\lambda) = -\Delta(\lambda)$, and Theorem 9 turns this into the statement that Δ has no nonzero *real* zero: such a zero would force $\sinh x = x$ with $x \neq 0$. That is what keeps c_* large. Break the symmetry and the obstruction vanishes: a real zero emerges from the origin at first order in $\theta - \frac{1}{2}$, and the counterexamples change character. For θ near $\frac{1}{2}$ the cheapest non-polynomial solution is the pure exponential $e^{\lambda_\theta x}$, with no oscillation at all, and it is cheap because it is a small perturbation of the constant solution. The oscillatory counterexamples, and the large threshold that comes with them, are specific to the midpoint.

What remains open is the global behaviour of c_* on $(0, 1) \setminus \frac{1}{2}$. One warning: this is harder to explore numerically than it looks. The zeros of G_θ are not confined to bands as in Theorem 12, and a search seeded from the asymptotic branches misses zeros. At $\theta = 0.35$, for instance, such a search returns the real zero -4.0399904 as the minimiser, whereas an argument-principle count over $|\operatorname{Re} \lambda| \leq 3.90$ reports four zeros rather than the two at the origin, and the extra pair is $3.8722154 \pm 11.6143549i$, which are the true minimisers. Any numerical study of $c_*(\theta)$ should be certified by contour counts and not by root finding alone.

Two scales at once

Suppose $f \in M_{h_1} \cap M_{h_2}$ with $h_1 \neq h_2$, and suppose f is of at most exponential growth. By Theorem 23 and the linear independence of distinct exponentials, f is a finite combination of modes $e^{\lambda x}$ with λ a characteristic root for *both* scales, that is with $\lambda \in \frac{2}{h_1}Z \cap \frac{2}{h_2}Z$ where $Z := E^{-1}(0)$. A nonzero common root would require $z/z' = h_1/h_2 > 0$ for two zeros z, z' of E , that is, two zeros of $\sinh z - z$ lying on a common ray from the origin. This reduces the two-scale rigidity problem to a concrete question.

Question 27. *Are any two distinct zeros of $\sinh z - z$ positively proportional? Equivalently, is $k \mapsto \arg z_k$ injective on $k \geq 1$?*

Numerically $\arg z_k$ is strictly increasing for $1 \leq k \leq 2000$, rising from 1.21706 toward $\pi/2$, which would give injectivity and hence: *any two distinct scales, together with any exponential growth bound, force a polynomial of degree at most two*. A proof should follow from sharp two-term asymptotics for x_k and y_k together with a finite check, but we have not carried this out. Note that no irrationality of h_2/h_1 is needed for the statement, only the injectivity above.

Several dimensions

The natural analogue in $\mathbb{R}^n$ replaces the midpoint mean value property by a spherical or ball mean value property at one fixed radius. The one-radius version of the mean value property for harmonic functions is a classical topic and the fixed-radius theory there is delicate; a comparison with Zalcman's framework for mean value characterizations [13] would be the place to start.

Sharp constants for higher modes

Theorem 20 says that the solution space jumps in dimension exactly at each α_k . It would be attractive to have a closed form, or at least a clean variational characterization, for α_1 . At present it is defined only as $2\operatorname{Re} z_1$ where z_1 is a root of a transcendental equation.

Reproducibility

The root finder, the argument-principle checks, the residual computation and the figure scripts are short and are provided as a computational supplement, so that every numerical assertion in this paper can be regenerated from scratch. Running the supplement reproduces Table 1 to the digits shown and all six figures.

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I thank everyone who confidently answered “parabolas” when I first asked Question 1; the unanimity of the wrong answer is what convinced me the question was worth writing up.

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